Introduction

The goal of this exercise is to get a better grip on the concept of Factorial Designs, their advantages, and their corresponding tests and statistical analyses. We study this in the context of an engineering experiment in aerodynamics, before proceeding to conduct our versions of such an experiment using paper helicopters! Finally, we implement the ANOVA (Analysis of Variance) framework from lecture for our collected data.

The Experiment

We first look at a recent real-world application of factorial designs in aerodynamics experiments from 2020, which investigates the effect of surface roughness and other environmental factors on airfoil performance. Read through the paper, trying to identify the details of the design such as the factors and levels, and the motivation for a full factorial design. Also try to interpret the results and understand their significance.

Answer the following questions in full sentences.

1. What is the main motivation for conducting a full factorial design in such a setting? What would changing one factor at a time fail to detect in this study?

2. Which of the factors are significant from the ANOVA analysis? Interpret what it means if two factors and their interaction are all significant? Are they independent?

Conducting Your Own Experiment

Conduct your own factorial design experiment now with paper helicopters! Fold the provided templates after adjusting for different wing heights and helicopter weights (via paper clips) to analyze whether they have a significant effect on the duration of flight. Choose the levels that you wish to study in your experiment, and feel free to add other factors to your design.

Keep the following questions in mind, and write answers explaining your choice in each.

3. List out the factors and levels that you plan on including in your design experiment. Guess whether these factors would increase or decrease the flight time.

4. How many runs or replicates are you planning to conduct for each combination of factors? Will different people drop the helicopter or time the experiment for each run? Would this be a factor or would it simply be different replicates?

5. Is the same person going to fold every helicopter? What would happen if you drew out the designs instead of using a template? Does it make your experiment more robust? What does it mean for power?

6. Where are you going to drop the helicopter from? Suppose your engineering experiment involved very expensive equipment (such as rockets or drones) and a large cost associated with each run. How would you decide the drop height?

7. Sketch the dataframe that will guide your experiment. Include the following columns: an ID to keep track of each measurement unit, the index of the replicate, each experimental factor that will be used and its level, an index to track the treatment group number (`d`), and the measured response.

8. Should the runs be conducted in the order you listed in your design matrix? What might happen if you run all "heavy" helicopters first and all "light" helicopters later?

Data Analysis

We will now analyze our own collected data. Ensure that you tabulate your data into a `.csv` file. This can be done by entering your data into Google Sheets or Microsoft Excel before saving as `.csv`.

As earlier, use the `readr` package, which is included inside the `tidyverse`. If you haven't installed the tidyverse before, you can do so by running `install.packages("tidyverse")` once.

```
# load tidyverse (includes readr for CSVs)
library(tidyverse)
```

```
# load your collected data into R

#my_data <- 
```

EDA and Scatterplots

9. Explore how your helicopters' flight times differed by wing width. Create a beehive plot using the `ggplot2` and `ggbeeswarm` packages in `R`. If you don't have these installed, install them by running `install.packages("ggplot2")` and `install.packages("ggbeeswarm")` once.

Stick to a single level of weight while making this for height, or add levels of weight by color coding them!

```
# Load necessary packages

library(ggplot2)
library(ggbeeswarm)

# Beeswarm Plot (Fill in)

# my_data |>
#   ggplot(aes(x = , y = )) +
#   geom_beeswarm(cex = 3, size = 3)
```

Next, calculate the group means. You may choose to this for any one factor or group by both. This requires the `dplyr` package, which you can install using `install.packages("dplyr")`. Also calculate the difference of each group mean with the overall mean. You may plot this to see if it is meaningful. Is it linear, non-linear etc?

```
### Load necessary packages

library(dplyr)

### Calculate Group Means

# group_means <- my_data |>
#   group_by() |>
#   summarize(avg = mean())
# group_means
```

```r
### Group Means - Overall Means

# overall_mean <- mean()

# effects_df <- group_means |>
#   mutate(effect = avg - overall_mean)

# effects_df$effect
```

```r
### Plot:

# ggplot(effects_df,
#        aes(x = width,
#            y = effect,
#            group = 1)) +
#   geom_point(size = 3) +
#   geom_line(linewidth = 1) +
#   geom_hline(yintercept = 0, linetype = "dashed") +
#   labs(x = "Wing Width",
#        y = "Difference from Overall Mean")
```

Randomization Based Inference

10. Next, conduct a randomization based test for the F-statistic to check whether the width of the wings turned out to be significant in determining flight times. Again, note that you will have to do this after fixing a level for weight.

For reference, see the Stat 158 F-inference slides page.

```r
### Selecting the weight factor

# my_data_subset <- my_data |>
#   filter(weight == "light")
```

```r
### Function to simulate under the Null

shuffle4_f <- function(y, z) {
  Z <- sample(z) # source of randomness

  my_data_subset <- data.frame(flight_time = y,
                               width = Z)
  
  avg_var <- my_data_subset |>
    group_by(width) |>
```

```
    summarize(var = var(flight_time)) |>
    summarize(avg_var = mean(var)) |>
    pull(avg_var)
  var_avg <- my_data_subset |>
    group_by(width) |>
    summarize(avg = mean(flight_time)) |>
    summarize(var_avg = var(avg)) |>
    pull(var_avg)

  var_avg / avg_var
}
```

```
### Calculating the observed value of F - statistic

avg_var <- my_data_subset |>
  group_by(width) |>
  summarize(var = var(flight_time)) |>
  summarize(avg_var = mean(var)) |>
  pull(avg_var)

var_avg <- my_data_subset |>
  group_by(width) |>
  summarize(avg = mean(flight_time)) |>
  summarize(var_avg = var(avg)) |>
  pull(var_avg)

f_stat <- var_avg / avg_var
```

```
### Plotting the Null Distribution

# null_stats <- replicate(500, shuffle4_f(y = , z = ))
null <- data.frame(stats = null_stats)

ggplot(null, aes(x = stats)) +
  geom_histogram() +
  geom_vline(xintercept = f_stat, color = "tomato", lwd = 2)

### Calculating p-value

null |>
  summarize(pval = mean(stats > f_stat))
```

Model Based Inference

11. We now move into two factor results using the Model Based approach.

What are the assumptions of the Model-based F-test used here?

Now, proceed to calculate the ANOVA Table as discussed in Lecture. Note that you can use `+` to add more than 1 factor, and `*` to specify interactions such as `aov(y ~ x1 + x2 + x1:x2)`.

```
### Calculating the ANOVA table

# helicopter_anova <- aov(, data = my_data)
# anova_table <- summary(helicopter_anova)
# anova_table
```

What factors do you find significant? Do they align with the EDA and the Randomization based tests?

Interaction Plots

12. Finally, we plot another simple diagnostic to check the pattern of any interaction between the two factors, wing width and helicopter weight (paper clips). We do this by plotting separate line graphs of flight time by width for both "light" and "heavy" helicopters.

If the lines are roughly parallel, this indicates that the effect of "heaviness" doesn't change with wing width. However if they aren't the plot gives you an idea of how this effect of weight changes with width, i.e it gives you an idea of how the interaction effect behaves.

```
### First compute group means for each combination
### Fill in appropriate column names

interaction_means <- my_data |>
  group_by(width, weight) |>
  summarize(avg_flight_time = mean(flight_time))

interaction_means
```

```
### Plot the interaction
### Fill in appropriate column names
```

```
ggplot(interaction_means,
       aes(x = width,
           y = avg_flight_time,
           group = weight,
           color = weight)) +
  geom_point(size = 3) +
  geom_line(linewidth = 1) +
  labs(x = "Wing Width",
       y = "Average Flight Time",
       color = "Paperclip Weight")
```

Are the two lines parallel? Is this consistent with the ANOVA table results above?

Answer in brief: how did the structure of your experiment affect what conclusions you were allowed to draw from its final results?